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UK Visa Status: Student Visa. Eligible for Graduate Visa (2 Years Full Working Rights)

STATEMENT

AI Engineer with 3 years of industry experience designing, building and deploying production **GenAI** applications on **LLMs** (**Azure OpenAI GPT-3.5**, self-hosted **Llama-3**) at an asset manager: **RAG** with **LlamaIndex** over **PostgreSQL** with **pgvector**, tool-calling **AI agents**, **prompt engineering** with guardrails and Git-versioned prompts, and **FastAPI** microservices on **AWS** with **Docker** and **Jenkins** CI/CD. Worked directly with Compliance, Client Service and Investment stakeholders to turn requirements into working AI tools, released through security testing (**VAPT**) and Compliance review; acted as de facto team lead for up to 4 interns and was twice recognised internally with Best Team Project (2024, 2025). Experienced with **Claude Code** as a development tool. Recently completed an MSc in Data Science at the University of Bristol, with a dissertation in partnership with AstraZeneca. Seeking a contract AI Engineer role building agents, RAG and AI-assisted engineering workflows for enterprise teams.

EDUCATION

University of Bristol – MSc. Data Science (awaiting result) September 2025 – September 2026
NMIMS B.Tech Information Technology (Dual Degree) June 2017 – April 2022
NMIMS MBA Technology Management (Dual Degree) June 2017 – April 2022

WORK EXPERIENCE

HDFC Asset Management Company Ltd. July 2022 – July 2025

AI Solutions Engineer / Forward Deployed Engineer

LLM Applications, Agents & RAG

- Designed and built **GenAI** applications end to end (**ReactJS** frontend, **FastAPI** backend, **RAG** logic, **pgvector** database and embeddings), starting as one of two people on the team, for Compliance, Client Service and Investment (**LlamaIndex** with **Azure OpenAI GPT-3.5 & Llama-3**); used department-based retrieval, with a separate **pgvector** database for each department so access stayed siloed.
- Built **AI agents** with **tool calling**: agents call **Python** functions as tools to run **SQL**-backed fund-comparison calculations and generate **PDF** comparison reports on user request.
- Built a **Python** extraction pipeline that turns over **300** unstructured daily emails from credit rating agencies into structured data with **Azure OpenAI**, fuzzy-matches them to tracked companies and sends analysts a daily summary, **saving 60 manual analyst hours a month**; it began on **AWS Textract** and moved to **Azure OpenAI** when a rating agency began sending emails as images, which also improved accuracy.
- Guardrailed responses through **system prompts**, versioned every prompt change in **Git**, and released to real users, changing the applications based on what broke; the **Compliance** department reviewed each application before release. Allocated models by cost: open-source **Llama-3** self-hosted on a **GPU EC2** instance for the largest department (Client Service) and stronger **Azure OpenAI** outputs for decision makers.

Enterprise Integration, Cloud & Secure Deployment

- Built the AI services as **Python FastAPI** microservices (**uvicorn**) in **Docker** behind a **ReactJS** frontend served by **Nginx**; designed the infrastructure on **AWS** using core services such as **EC2, S3, ECR** and **Lambda**, with separate frontend and backend repositories and containers and a one-click **Jenkins** and **GitLab CI/CD** pipeline for dev and prod.
- Implemented **RBAC** for the **GenAI** apps with a Python authentication function against the company **LDAP** server alongside department-level access to documents; restricted access to company private IPs using **AWS Security Groups**; deployed an **Nginx** reverse proxy, one of the fixes that took the applications from repeatedly failing to

passing the **VAPT** (vulnerability assessment and penetration testing); coordinated with the security team on port whitelisting and development safeguards.

Stakeholders, Data & Team

- Designed a **Normalized SQL Schema** (**PostgreSQL**) and wrote **Python** ingestion scripts for the **Morningstar API** covering every fund in the Indian mutual fund market (2023–2025), integrated with legacy data from analysts' Excel files; developed **Optimized SQL** queries, **SQL Functions** and **Procedures**.
- Developed **Power BI** dashboards for fortnightly credit report analysis, migrating financial analysts from Excel to Power BI and improving analyst efficiency by **10 hours a week**; then recreated the dashboards in the **ReactJS** frontend so the wider organisation could use the data without Power BI licensing costs.
- Worked directly with users in Client Service, Compliance and Investment to turn their requirements into working tools that replace manual work; acted as **de facto team lead**, guiding up to 4 interns in my final year; recognised with **Best Team Project** twice (2024 rating-email automation, 2025 department RAG chatbot).

PROJECTS

- **University of Bristol MSc. Data Science Dissertation in partnership with AstraZeneca:** CellLineSelector (Team project, co-built; I designed the RAG-based AI assistant and worked on the scoring algorithm and data preprocessing for a tool, built as FastAPI microservices with a React and TypeScript frontend, that ranks cancer cell lines by target inclusion and exclusion genes and returns the evidence behind each pick; validated and refined with feedback from AstraZeneca scientists.) GitHub: https://github.com/Arishtanemi4/az-team25.git | Demo: http://celllineselector-frontend-408937187029.s3-website.eu-west-2.amazonaws.com/
- **Wingman:** AI dating-coach web app I built as a side project with 108 LLM personas across 12 archetypes (ReactJS, Python FastAPI, Kimi K2 and DeepSeek V4 Pro via NVIDIA NIM). Behaviour steered by prompts; LLM output treated as untrusted: validated with Pydantic, repaired and cached by input hash; rate limits shaped the design (12 calls instead of 108, serialised calls, results streamed to the UI). GitHub: https://github.com/Arishtanemi4/wingman.git
- **Pauti:** offline-only expense-splitting Android app on a CRDT architecture. GitHub: https://github.com/Arishtanemi4/pauti.git **| Contact Reminder:** Android app that texts contacts on birthdays and anniversaries. GitHub: https://github.com/Arishtanemi4/contact-reminder.git
- **University of Bristol MSc Applied Data Science Group Project:** VisionCaptioner (Team project. Architecture study comparing CNN/ViT/ResNet encoders and LSTM/attention decoders, benchmarked against BLIP/SigLIP and commercial VLMs.) GitHub: https://github.com/Arishtanemi4/ata-team30.git
- **B.Tech Final Year Project:** Udacity self-driving car simulator. (A self-driving car trained using a **Convolution Neural Network (CNN)** model.)

ACHIEVEMENTS

- **Project Hack 27**: Rolls Royce Derby 4th Place winner. (Human Centric Data Analytics tool to predict the competition winner based on human wellbeing and psychological temperament survey data.) | GitHub: https://github.com/Arishtanemi4/team-4b.git
- **Encode Vibe Coding Hackathon:** Codeplain.ai track 2nd place winner. (Wingman.ai – An AI powered dating app simulation using codeplain's spec-driven framework) | GitHub: https://github.com/Arishtanemi4/wingman-codeplain.git
- **University of Bristol Datathon sponsored by Lloyd's Bank:** 3rd Place winner. (Kaggle Credit card fraud detection problem.)

SKILLS

- **Tools:** Claude Code, Azure OpenAI, LlamaIndex, Jupyter, Anaconda, AWS (EC2, S3, ECR, Lambda, SageMaker), Docker, Jenkins, GitLab, Git, Nginx, Linux, Power BI, VS Code.
- **Languages:** Python (pandas, scikit-learn, FastAPI, Pydantic), SQL, R (dplyr, tidyverse, ggplot2), TypeScript and NodeJS (express, ReactJS), Shell
- **Databases:** PostgreSQL (pgvector, SQL functions and procedures), SQLite